SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: CSA16 **COURSE CODE:** Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3, BL4, BL5)

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Experiments

Criteria	**Understanding of Concepts (2.5)**	**Experimental Design (3)**	**Use of Tools (2)**	**Analysis of Results (1.5)**	**Documentation (1)**	**Total (10)**
Weightage	25%	30%	20%	15%	10%	**100%**
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

I **Kongara Jogesh (192525035)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Kongara Jogesh

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Annexure C

Experiments – Solutions

Question 1 (10 Marks)

Children of three ages are asked to indicate their preference for three photographs of adults (A, B, C). We test whether there is a significant relationship between the age of the child and photograph preference using the Chi-Square Test of Independence.

Observed Data:

Age of child	Photograph A	Photograph B	Photograph C	Row Total
5–6 years	18	22	20	60
7–8 years	2	28	40	70
9–10 years	20	10	40	70
Column Total	**40**	**60**	**100**	**200**

Step 1: Hypotheses

H0: Age of the child and photograph preference are independent (no relationship).

H1: Age of the child and photograph preference are not independent (a relationship exists).

Step 2: Expected Frequencies

Expected value = (Row Total × Column Total) / Grand Total

Age of child	Photograph A	Photograph B	Photograph C
5–6 years	12.0	18.0	30.0
7–8 years	14.0	21.0	35.0
9–10 years	14.0	21.0	35.0

Step 3: Chi-Square Statistic

$\chi^2 = \Sigma [(O - E)^2 / E]$

Calculated $\chi^2 = 29.60$, Degrees of freedom = $(3-1)(3-1) = 4$, p-value ≈ 0.0000058 (< 0.05)

Since the p-value is far less than 0.05 (and $\chi^2 = 29.60$ exceeds the critical value 9.488 at $\alpha = 0.05$, df = 4), we reject H0. There IS a statistically significant relationship between the age of the child and photograph preference.

What is wrong with this study?

• Sample size imbalance / cell count concerns: some expected cell frequencies are close to the recommended minimum of 5, which can make the chi-square approximation less reliable.

• No random sampling described: it is not stated how the children were selected, raising concerns about selection bias and generalisability of results.

• Confounding variables ignored: factors such as gender, cultural background, or familiarity with the photographed adults are not controlled for.

• Nominal treatment of an ordinal variable: age is naturally ordinal, but the chi-square test treats it as purely categorical/nominal, discarding information about the ordering of age groups (an ordinal test such as the Cochran–Armitage trend test would use this information better).

• Causation cannot be inferred: a significant association only shows correlation between age and preference, not that age causes the preference.

R Code:

```
# Preference counts for photographs A, B, C across age groups
A <- c(18, 2, 20)
B <- c(22, 28, 10)
C <- c(20, 40, 40)
pref <- data.frame(A, B, C)

# Chi-square test of independence
mat <- matrix(c(A, B, C), nrow = 3)
chisq.test(mat)

# Sample covariance between B and C
cov(B, C)

# Sample covariance matrix
cov(pref)

# Sample correlation between B and C
cor(B, C)

# Sample correlation matrix
cor(pref)
```

Output – Covariance:

cov(B, C) = -20.00

	A	B	C
A	97.33	-74.00	-46.67
B	-74.00	84.00	-20.00
C	-46.67	-20.00	133.33

Output – Correlation:

cor(B, C) = -0.189

	A	B	C
A	1.000	-0.818	-0.410
B	-0.818	1.000	-0.189
C	-0.410	-0.189	1.000

Interpretation: B and C are weakly negatively correlated (-0.189), meaning children who preferred photograph B slightly tended not to prefer photograph C, and vice-versa. A and B show a strong negative correlation (-0.818), suggesting that as preference for A rises across age groups, preference for B tends to fall.

Question 2 (10 Marks)

To visualise the distribution of points scored by the 15 players and identify scoring outliers, a boxplot (box-and-whisker plot) is the most appropriate tool. A boxplot summarises the five-number summary (minimum, Q1, median, Q3, maximum) and flags any value beyond 1.5×IQR from Q1/Q3 as a potential outlier.

Player ID	Player Name	Points Scored
P1	Player A	12
P2	Player B	15
P3	Player C	14
P4	Player D	10
P5	Player E	18
P6	Player F	20
P7	Player G	22
P8	Player H	13
P9	Player I	11
P10	Player J	35
P11	Player K	16
P12	Player L	17
P13	Player M	19
P14	Player N	21
P15	Player O	23

Step 1: Sort the data

10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 35

Step 2: Five-Number Summary

Minimum = 10

Q1 (25th percentile) = 13.5

Median (Q2) = 17

Q3 (75th percentile) = 20.5

Maximum = 35

IQR = Q3 − Q1 = 20.5 − 13.5 = 7

Step 3: Outlier Fences

Lower Fence = Q1 − 1.5×IQR = 13.5 − 10.5 = 3.0

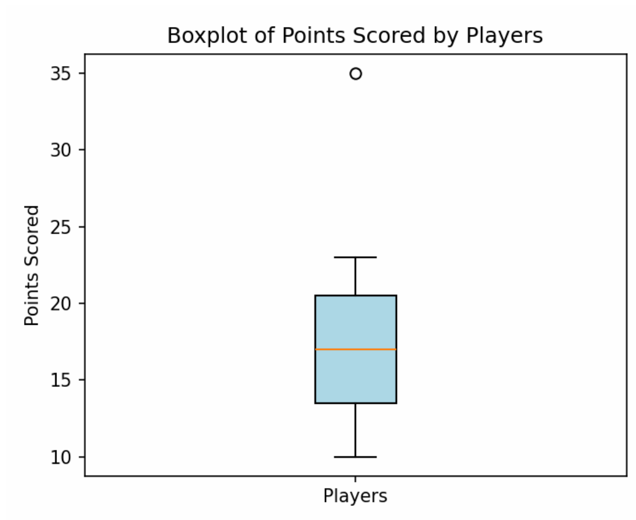
Upper Fence = Q3 + 1.5×IQR = 20.5 + 10.5 = 31.0

Any data point below 3.0 or above 31.0 is flagged as an outlier. Player J scored 35 points, which is above the upper fence of 31.0 — this is a statistical outlier and should be visualised as a separate point beyond the whisker on the boxplot.

R Code:

```
points <- c(12,15,14,10,18,20,22,13,11,35,16,17,19,21,23)
summary(points)
IQR(points)
boxplot(points,
        main = 'Boxplot of Points Scored by Players',
        ylab = 'Points Scored', col = 'lightblue')
```

Boxplot:



Inference: The distribution is fairly symmetric between 10 and 23, with the median at 17 points. Player J's score of 35 stands out clearly as an outlier — the coach should investigate whether this was an exceptional performance or a data-entry/measurement anomaly.

Question 3 (10 Marks)

We build a Decision Tree classifier (ID3 algorithm, using Information Gain based on Entropy) to predict Loan Approval (Yes/No) from Age, Income, Employment Status, and Credit Score.

Dataset:

#	Age	Income	Employment	Credit Score	Approval
1	young	high	employed	good	yes
2	young	high	self-employed	average	yes
3	middle	high	employed	good	yes
4	old	medium	employed	good	yes
5	old	low	unemployed	poor	no
6	old	low	self-employed	average	no
7	middle	low	unemployed	poor	no
8	young	medium	employed	average	yes
9	young	low	unemployed	poor	no
10	old	medium	self-employed	good	yes
11	middle	medium	employed	average	yes
12	middle	high	self-employed	good	yes
13	old	medium	unemployed	poor	no
14	young	medium	self-employed	average	yes
15	middle	low	employed	poor	no

Step 1: Entropy of the Target (Approval)

Out of 15 records: 9 = Yes, 6 = No

Entropy(S) = −(9/15)log2(9/15) − (6/15)log2(6/15) = 0.971

Step 2: Information Gain for Each Attribute

Information Gain = Entropy(parent) − Weighted Average Entropy(children)

Attribute	Information Gain
Age	0.083
Income	0.711
Employment	0.470
Credit Score	**0.730 (highest)**

Credit Score gives the highest information gain (0.730), so it is selected as the Root Node.

Step 3: Split on Credit Score

• Credit Score = good → 5/5 records = Yes → Pure leaf: Approval = YES

• Credit Score = poor → 5/5 records = No → Pure leaf: Approval = NO

• Credit Score = average → mixed (4 Yes, 1 No) → needs further splitting

Step 4: Split the 'average' Credit Score branch

Within Credit Score = average, computing information gain for the remaining attributes shows Age gives a perfect (pure) split (Gain = 0.722):

• Age = young → 3/3 = Yes → YES

• Age = middle → 1/1 = Yes → YES

• Age = old → 1/1 = No → NO

Final Decision Tree:

Credit Score

```
├── good      → Approval = YES
├── poor      → Approval = NO
└── average   → Age
      ├── young  → Approval = YES
      ├── middle → Approval = YES
      └── old    → Approval = NO
```

R Code:

```
library(rpart)
library(rpart.plot)

loan <- data.frame(
  Age = c('young','young','middle','old','old','old','middle','young',
          'young','old','middle','middle','old','young','middle'),
  Income = c('high','high','high','medium','low','low','low','medium',
             'low','medium','medium','high','medium','medium','low'),
  Employment = c('employed','self-employed','employed','employed','unemployed',
                 'self-employed','unemployed','employed','unemployed','self-employed',
                 'employed','self-employed','unemployed','self-employed','employed'),
  CreditScore = c('good','average','good','good','poor','average','poor','average',
                  'poor','good','average','good','poor','average','poor'),
  Approval = c('yes','yes','yes','yes','no','no','no','yes','no','yes',
               'yes','yes','no','yes','no')
)

tree_model <- rpart(Approval ~ Age + Income + Employment + CreditScore,
                    data = loan, method = 'class')
rpart.plot(tree_model, type = 4, extra = 104)
```

Interpretation: Credit Score is the strongest predictor of loan approval — applicants with a 'good' score are always approved and those with a 'poor' score are always rejected. For applicants with an 'average' credit score, Age becomes the deciding factor, with only 'old' applicants being rejected.

Question 4 (5 Marks)

Class A: 76, 35, 47, 64, 95, 66, 89, 36, 84

Class B: 51, 56, 84, 60, 59, 70, 63, 66, 50

(i) Mean, Median and Range

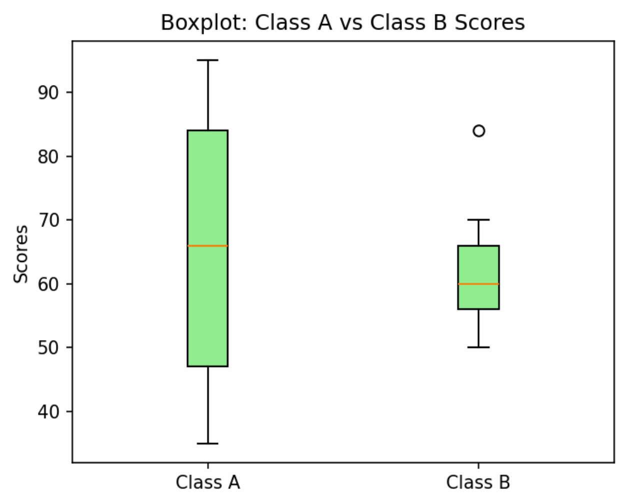
Statistic	Class A	Class B
Mean	65.78	62.11
Median	66.00	60.00
Minimum	35	50
Maximum	95	84
Range	60	34

Class A has a higher mean (65.78 vs 62.11) and a higher median (66 vs 60). Class A also has a much larger range (60 vs 34), indicating far greater spread/variability in scores.

(ii) Boxplot and Inference

R Code:

```
ClassA <- c(76,35,47,64,95,66,89,36,84)
ClassB <- c(51,56,84,60,59,70,63,66,50)
boxplot(ClassA, ClassB, names = c('Class A','Class B'),
        col = c('lightgreen','lightyellow'),
        main = 'Boxplot: Class A vs Class B Scores', ylab = 'Scores')
```



Inferences:

• Class A shows a wider spread of scores (IQR and whiskers longer), meaning student performance was less consistent — some students scored very low (35–36) while others scored very high (89–95).

• Class B's scores are more tightly clustered (50–84), indicating more consistent performance across students, despite a slightly lower average.

• Class A's higher median suggests that, typically, Class A students performed better, but with higher risk/variance, whereas Class B shows more uniform, moderate performance.

Question 5 (10 Marks)

Age and body-fat percentage (%fat) were recorded for 18 randomly selected adults:

Age	23	23	27	27	39	41	47	49	50
%fat	9.5	26.5	7.8	17.8	31.4	25.9	27.4	27.2	31.2

Age	52	54	54	56	57	58	58	60	61
%fat	34.6	42.5	28.8	33.4	30.2	34.1	32.9	41.2	35.7

(a) Mean, Median, and Standard Deviation

Statistic	Age	%Fat
Mean	46.44	28.78
Median	51.00	30.70
Standard Deviation	13.22	9.25

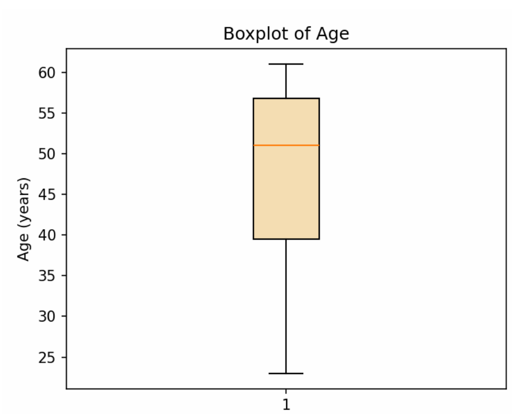
R Code:

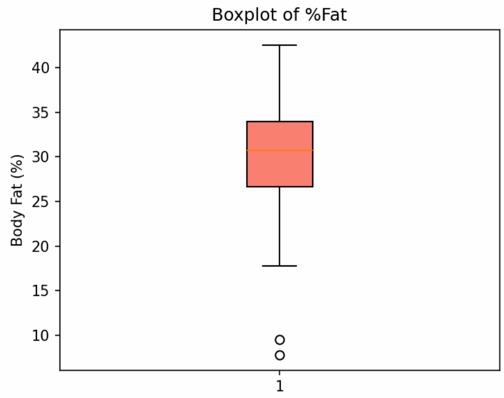
```
age <- c(23,23,27,27,39,41,47,49,50,52,54,54,56,57,58,58,60,61)
fat <- c(9.5,26.5,7.8,17.8,31.4,25.9,27.4,27.2,31.2,
         34.6,42.5,28.8,33.4,30.2,34.1,32.9,41.2,35.7)

mean(age); median(age); sd(age)
mean(fat); median(fat); sd(fat)
```

(b) Boxplots for Age and %Fat

```
boxplot(age, main = 'Boxplot of Age', ylab = 'Age (years)', col = 'wheat')
boxplot(fat, main = 'Boxplot of %Fat', ylab = 'Body Fat (%)', col = 'salmon')
```



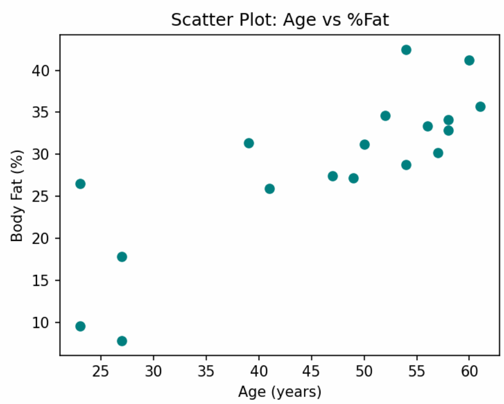


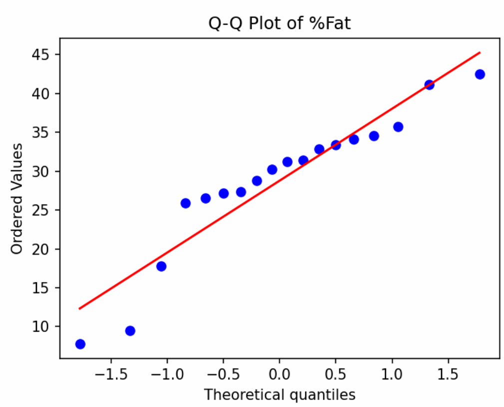
Both distributions are roughly symmetric with no extreme outliers; %fat shows slightly more spread relative to its scale than age.

(c) Scatter Plot and Q–Q Plot

```
plot(age, fat, main = 'Scatter Plot: Age vs %Fat',
     xlab = 'Age (years)', ylab = 'Body Fat (%)', pch = 19, col = 'darkblue')
cor(age, fat)

qqnorm(fat, main = 'Q-Q Plot of %Fat')
qqline(fat, col = 'red')
```





Correlation coefficient (age vs %fat) = 0.818

Inferences:

• The scatter plot shows a strong positive linear relationship between age and body fat percentage ($r = 0.818$) — as age increases, %fat tends to increase as well.

• The Q-Q plot shows the %fat data points lying reasonably close to the reference line, suggesting the distribution of %fat is approximately normal, with only minor deviation at the tails.